

**VIT**Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

REG.NO.:

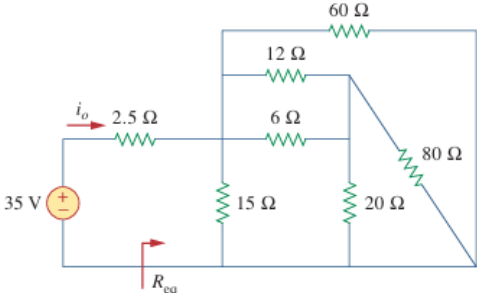
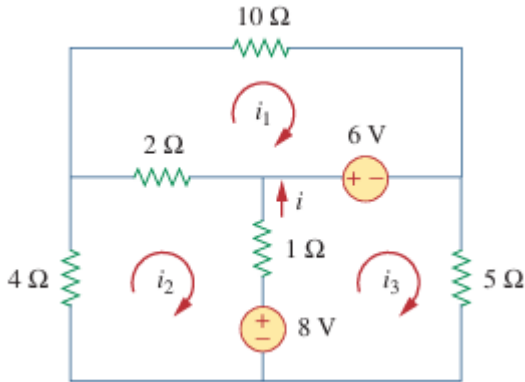
SCHOOL OF ELECTRONICS ENGINEERING
CONTINUOUS ASSESSMENT TEST - I
FALL SEMESTER 2025-2026

SLOT: C2 +TC2+TCC2

Programme Name & Branch : B. Tech BVD
 Course Code and Course Name : BAECE103 – Network Theory
 Faculty Name(s) : Dr. Elizabeth Rufus
 Class Number(s) : VL2025260105993
 Date of Examination : 19.08.2025
 Exam Duration : 90 minutes Maximum Marks: 50

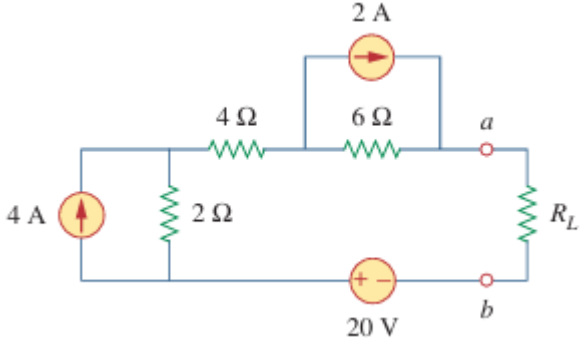
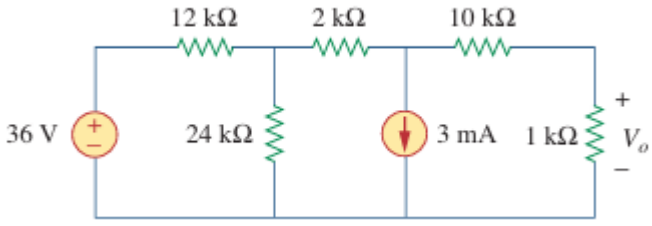
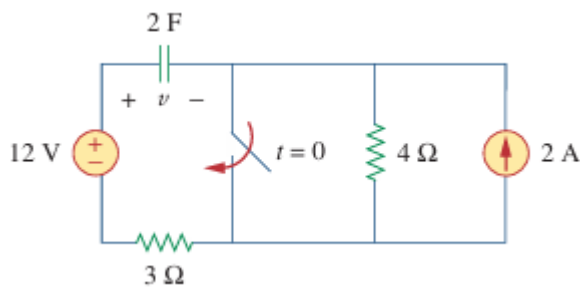
General instruction(s):

- Answer All Questions.
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyze, 5 – Evaluate, 6 – Create)
- Course Outcomes:
CO1: Compute DC circuit parameters using circuit laws and theorems.
CO2: Determine the natural and step responses of RC, RL, and RLC circuits.

Q. No	Question	M	CO	BL
1.	Obtain the equivalent resistance in the circuit of Fig. 1.  Fig.1	10	1	2
2.	Using mesh analysis, find i in the circuit of Fig.2  Fig.2	10	1	3



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3.	For the circuit in Fig .3, find R_L for maximum power deliverable to the load and also find the maximum power.  Fig.3	10	1	3
4.	Apply Norton's theorem to find V_o in the circuit of Fig.4.  Fig.4	10	1	3
5.	Calculate the capacitor voltage for $t < 0$ and $t > 0$ in the circuit of Fig. 5.  Fig.5	10	2	3